Project #2: Astronomy/Exoplanet Research

Question: Can small deviations in an exoplanet's transit times reveal gravitational interactions with other planets, and can a data-driven model distinguish genuine transit-timing variations from observational noise?

Imagine a planet transiting its star every 10 days.
If everything were perfectly Keplerian:

Transit 1: day 0
Transit 2: day 10
Transit 3: day 20
Transit 4: day 30

But if another planet is gravitationally influencing it, you might observe:

0.00
10.03
20.01
29.96
40.02

Those tiny deviations are transit-timing variations (TTVs).
What does the departure from a perfectly periodic orbit tell us about the architecture of the planetary system?

Skills to Learn: stellar photometry, transit geometry, orbital periods, transit depth, signal-to-noise, light curves, detrending, cadence, limb darkening, TESS systematics, uncertainty estimation

Python pipeline: Target selection -> TESS data retrieval -> Quality filtering -> Detrending -> Transit identification -> Transit-by-transit fitting -> Mid-transit time extraction -> O-C diagram -> TTV analysis

Dive deeper: Which observable properties of planetary systems are associated with stronger and more detectable TTV signals?

Possible variables: orbital period, planet radius, planet mass, period ratio, stellar type, orbital separation, multiplicity, proximity to orbital resonances

Statistical Analysis: TTV amplitude ~ f(period ratio, planet radius, stellar properties,...)
Don't stop at Pearson correlation compare Pearson, Spearman, Robust Regression, Bootstrap Confidence Intervals